Supplementary Material

HyperDG: Hyperbolic Representation Alignment for Robust Domain Generalization via Curvature Refinement

Sunny Gupta Amit Sethi

1 Overview

This supplementary document accompanies the revised submission and contains additional visualizations, expanded ablations, and implementation details requested during review. In particular, we include:

- Poincaré disk visualizations across all experimental settings.
- Curvature evolution plots during training.
- Single-curvature vs per-domain curvature comparisons.
- MERU backbone supplementary results.
- Runtime and implementation notes.

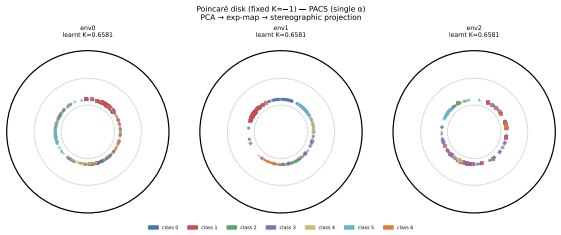
2 Poincaré Disk Visualizations

We project learned Lorentz embeddings into the Poincaré disk using:

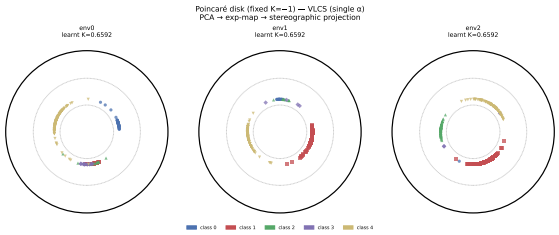
1. Tangent-space extraction,
2. PCA reduction to 2D,
3. Exponential map with fixed curvature $K = -1$,
4. Stereographic projection.

These figures qualitatively illustrate the geometric structure learned by HyperDG.

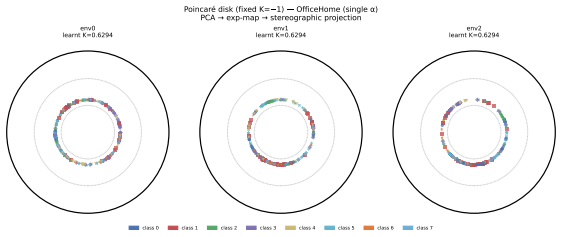
2.1 HyperDG (Per-Domain Curvature)



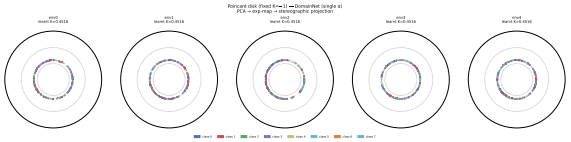
(a) PACS



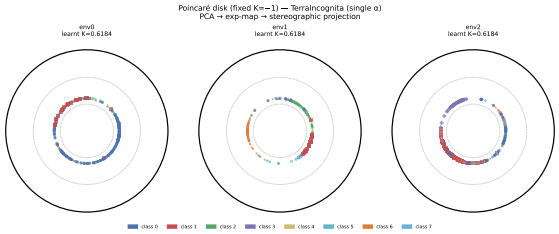
(b) VLCS



(c) OfficeHome



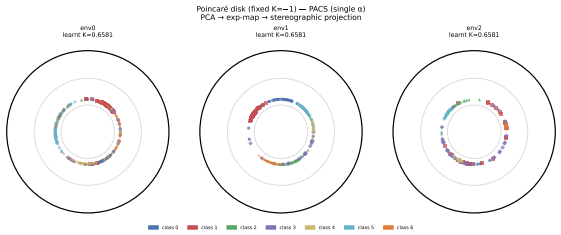
(d) DomainNet



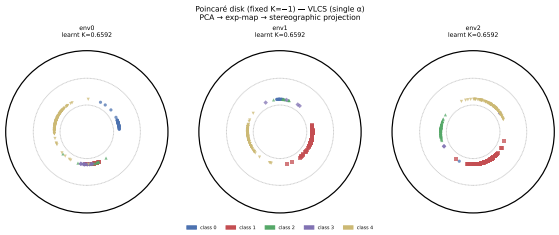
(e) DomainNet

Figure 1: HyperDG using per-domain learnable curvatures. Distinct radial cluster depths indicate geometry specialization across domains.

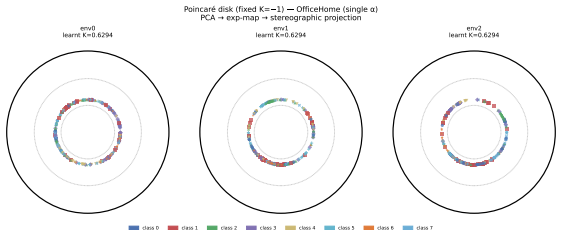
2.2 HyperDG (Single Shared Curvature)



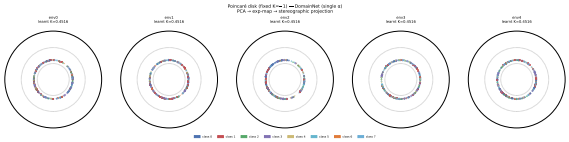
(a) PACS



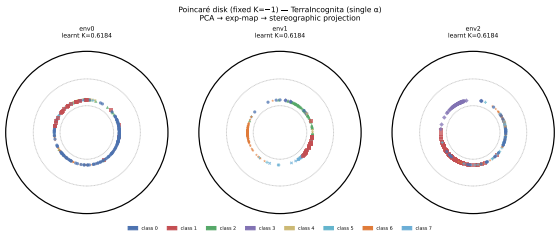
(b) VLCS



(c) OfficeHome



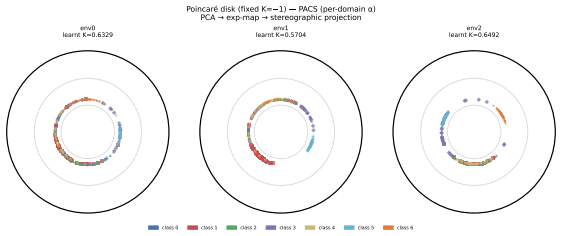
(d) DomainNet



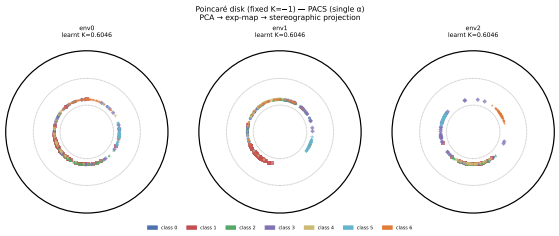
(e) TerraIncognita

Figure 2: HyperDG using a single shared learnable curvature. Radial distributions become more uniform.

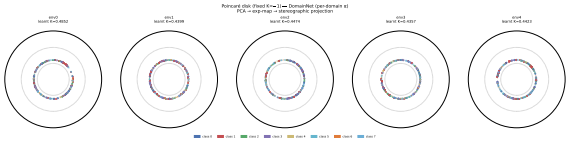
2.3 MERU Backbone Visualizations



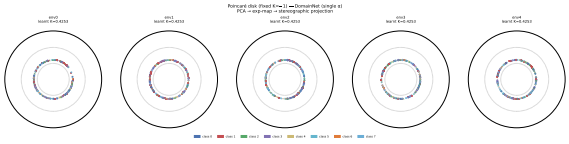
(a) PACS + Per-domain



(b) PACS + Single Curvature



(c) DomainNet + Per-domain



(d) DomainNet + Single Curvature

Figure 3: MERU-based variants exhibit tighter clusters due to stronger hyperbolic priors from pretraining.

3 Additional Ablation Tables

3.1 Single vs Multiple Curvature

Table 1: Comparison between per-domain and single shared curvature.

Dataset	Per-domain	Single Shared
PACS	92.2	92.2
VLCS	81.8	81.5
OfficeHome	76.8	76.6
DomainNet	52.1	52.6
TerraIncognita	53.1	53.6

3.2 Runtime Overhead

Table 2: Wall-clock training cost.

Method	ms/step	Relative Cost
ERM	464.81	1.00×
SFT	581.40	1.25×
HyperDG	458.69	0.99×

4 Implementation Notes

All supplementary experiments use the same optimizer, seeds, splits, and evaluation protocol as the main paper unless otherwise stated.

5 Conclusion

We thank the reviewers for their constructive feedback. These additional results further support the robustness, efficiency, and geometric consistency of HyperDG.